Ballast — Presenter Guide

Everything you need to present, demo and defend Ballast in 15 minutes. No finance background is assumed.
Team globalhost · live at **ballast.jkartik.in**

> **The whole pitch in one breath (memorise this):**
>
> "A bank must pay every customer withdrawal instantly, but cash that sits idle earns nothing. Every evening at 5 pm it can lock spare cash into an overnight loan that earns interest, but that money is frozen until 8 am. Lock too much and you can't pay customers; lock too little and you waste millions. Ballast decides, every night, exactly how much to keep back. On a full year of data it cut the cost of liquidity by 51%, about $21 million, never missed a single payout, and came within 3% of a strategy that could see the future."

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1 · Finance crash course (10 minutes of reading)

Read this section twice. It is all the finance you need. Every term the judges might use is here, in plain English.

The situation

Imagine a payments company or bank holding about **$2 billion** of customers' money. Every hour, money comes in (deposits) and goes out (withdrawals): around **$10 million each way per hour**. The company must pay every withdrawal immediately. That is the one rule that can never break.

Cash sitting in the account earns **0%**. That is "idle cash", and the problem is called **The Reserve Juggler: the Idle Cash Problem**.

```
 5:00 pm                                                    8:00 am
   |---------------- 15 hours: the repo is LOCKED -------------|
   ▲                                                          ▲
SWEEP: move spare cash into overnight repo          repo matures, cash + interest come back
(earns ~4% a year)                                   (customers kept withdrawing all night!)

Customers keep withdrawing overnight → you must keep a BUFFER of cash back at 5 pm.
```

TERM	WHAT IT MEANS (SAY IT LIKE THIS)
Tier 0 / cash	Money in the account. Instantly available, earns nothing.
Repo / Tier 1	An overnight loan you make to a safe counterparty. Earns about **4% per year**. You put money in at 5 pm and get it back at 8 am. "The yield vault."
Sweep	The 5 pm move of spare cash into repo. Ballast's whole job is deciding the sweep size.
Buffer	The cash you deliberately *don't* sweep, kept to pay customers overnight.

Early-break fee	If you run out of cash overnight, you can pull money out of repo early, but it costs a **0.5% penalty**. That sounds small, but for a 15-hour loan it equals a **292% annual rate** (0.5% × 8,760 h ÷ 15 h). Wiping out the interest from months of good nights takes only one break.
Overdraft	Emergency borrowing from the bank at **12% a year**. The absolute last resort.
Bonds (Tier 2) / Equities (Tier 3)	Other places to park money. They pay more, but selling them quickly costs 2% (bonds) or 15% (stocks). Ballast does not use them (see Trade-offs).
Funding waterfall	The order in which money is found if cash runs out: break repo, then sell bonds, then sell stocks, then overdraft. This is why **payouts never fail**.
Total cost of liquidity	The score we minimise: interest lost on idle cash plus every fee and penalty paid. Lower is better.
Worst overnight drain ("path-max", M)	The deepest point that cumulative withdrawals reach during the night. Not the total at 8 am, the *lowest point*. Like a road trip: your fuel tank must survive the worst stretch, not just the final distance. This is our key insight.
Percentile / quantile	"The 99.5th percentile drain" = the drain that 99.5% of past nights stayed under.
Newsvendor model, q*	A classic formula for "how much to stock when demand is uncertain" (like a newspaper seller). Cost of being short = 0.5% break fee. Cost of being over = 15 hours of lost interest ≈ 0.007%. Being short is ~70× worse, so you should cover about **98.6%** of nights. That is q*.
Coverage floor (99.5%)	A safety rule we add on top: always cover at least 99.5% of nights, even though the maths says 98.6% is enough. Stricter is safer.
Regime (weekday vs weekend)	Friday and Saturday nights drain about **5× more** than other nights. So Ballast keeps two separate histories and two buffers.
Baseline	A simpler strategy we compare against: all-cash (never sweep), fixed buffer, moving average + 2σ, Miller–Orr (a textbook cash model).
Oracle	A cheating strategy that knows the future. Not usable. It shows the best score theoretically possible, so "how close are we to perfect?"
Look-ahead / leakage	Using future data by accident. Judges hate it. Ballast only uses data up to 4:59 pm on the day it decides, and the app *proves* it on every run.
Regret	The gap between Ballast and the oracle, split into causes (mostly forecast error).
bps (basis points)	Hundredths of a percent. 50 bps = 0.5%.
Stress test	Replay the year with a disaster injected (bank run, holiday rush) and see if payouts still happen.

How Ballast decides, in 5 steps (every night at 5 pm)

1. **Measure** every past night's worst drain.
2. **Split** nights into weekday and weekend, because weekends are about 5× riskier.
3. **Price** it: short costs 0.5%, idle costs about 0.007%, so cover about 98.6% of nights (raised to the 99.5% floor).

4. **Hold** that percentile of past drains as the buffer: about **$19M on weeknights, $47M on weekends**.
5. **Sweep** everything else into repo. If a night is worse than expected, the waterfall pays customers anyway.

Why it's smart: it is simple, explainable on a spreadsheet, fast, and it adapts to each night's real risk instead of using one fixed cushion.

2 · Numbers to memorise (reference dataset)

WHAT	NUMBER	HOW TO SAY IT
Total cost, Ballast	`$20.35M`	"Twenty point three million for the full year."
Total cost, all-cash (do nothing)	`$41.33M`	"Doing nothing costs forty-one million."
Saving vs all-cash	`$20.98M · 51%`	"We cut the cost in half: twenty-one million saved."
Saving vs Miller–Orr (textbook)	`$5.18M · 20%`	"Twenty percent better than the textbook model."
vs best fixed buffer ($40M, chosen with hindsight)	`$28K better`	"We even beat a buffer that was tuned by looking at the answers."
Gap to the oracle (perfect future knowledge)	`$598K · ~3%`	"Within 3% of perfect."
Net interest earned	`$21.35M`	
Payout success	`100%`	"Not one customer payment failed."
Break-even coverage q*	`98.63%`	Our floor is 99.5%.
Buffers	`$19.0M / $46.7M`	Weeknight / weekend, out of roughly $1–2B in cash.
Weekend vs weekday risk	`~5×`	Saturday drain is about 6.8× Sunday's.
Coverage achieved out of sample	`96.5%`	272 of 282 nights. The 10 misses were paid by breaking repo; payouts were still 100%.
Capital-constrained nights	`66 of 364`	Nights when the company simply didn't have enough cash. Even the oracle can't fix these.
Book drain over the year	`−$2.53B`	Withdrawals beat deposits by $2.5B, so by November the money runs out for *every* strategy. That is a capital problem, not a sweep problem.
Full-year simulation time	`~6 s`	8,760 hourly decisions.
Bonds tested at 5–50%	`−$75K to −$766K`	Every bond allocation lost money, so we switched bonds off.

3 · Who does what (3 speakers, 15 minutes)

Pick the most confident speaker as **A** (opens and closes), the one most comfortable with the laptop as **B** (method and demo), and the calmest under questions as **C** (results, honesty, Q&A lead).

`0:00–1:00`	**A**	Hook: the one-breath pitch above. Slide 1 (title).

1:00–3:30	A Slide 2, problem framing: the 5 pm decision, the 292% trap, weekends are 5× riskier.
3:30–6:00	B Slide 3, model: worst-drain idea, newsvendor q*, two regimes, waterfall. Slide 4 in 30 s (edge cases).
6:00–10:30	B **LIVE DEMO** (run-sheet in section 5). C stands next to the screen and points at numbers.
10:30–12:30	C Slide 6 results, then Slide 7 trade-offs (why no bonds, why a simple model).
12:30–14:00	C Slide 8, limits and next steps. Honesty wins trust.
14:00–15:00	A Close (Slide 9): three takeaways, invite them to scan the QR and try it. "Questions?"

Hand-offs: end your part by naming the next speaker and topic: "Now *Name* will show you how Ballast actually decides." It looks rehearsed and professional.

4 · The script, slide by slide

Don't read it word for word. Learn the **bold idea** of each paragraph and say it your way. The quoted lines are the strongest phrasings; keep those.

A Slide 1 — Title (1 min)

> "Good morning. We're team globalhost, and this is Ballast. [one-breath pitch]. Everything you'll see is live at ballast.jkartik.in. You can open it on your phone right now."

A Slide 2 — Problem framing (2.5 min)

- **The decision:** "Every day at 5 pm the treasury chooses how much cash to lock into overnight repo until 8 am. That's the only decision, and it repeats 365 times a year."
- **The trap:** "If you keep too little back and run short, breaking the repo costs 0.5%. For a 15-hour loan that is a 292% annual rate. One bad night can wipe out the interest from months of good ones."
- **The waste:** "Keep too much back and billions sit idle earning nothing. Doing nothing costs this company $41 million a year."
- **What we found in the data:** "Risk isn't the same every night. Friday and Saturday nights drain about five times more. Any single fixed cushion is wrong five nights a week."
- **The rule we never break:** "Every customer payout must be made on time. That's a constraint, not a goal."

> "So the question is: how much cash should we hold back tonight, so that nobody waits and nothing sits idle?" → hand to B.

B Slide 3 — Model and algorithm (2 min)

- **Insight 1, measure the right thing:** "Most approaches look at how much leaves overnight in total. But you run out of cash at the *worst moment* of the night, not at the end. We measure that worst moment for every past night."
- **Insight 2, price it:** "Running short costs 0.5%. Holding an idle dollar for 15 hours costs about 0.007%. Being short is about 70 times worse, so the classic newsvendor formula says to cover 98.6% of nights. We add a stricter 99.5% safety floor."
- **Insight 3, two regimes:** "Weekday and weekend nights each get their own history, so the buffer is about $19 million on weeknights and $47 million on weekends."
- **Safety net:** "If a night is still worse, a funding waterfall pays customers: break repo, then bonds, then stocks, then overdraft. That's why payouts are 100%."
- **No cheating:** "Each decision uses only data available before 5 pm, and the app re-proves that on every run."

B Slide 4 — Edge cases (30 s, then go to the demo)

> "Spikes are explained automatically: paydays and weekend spending. We stress-test bank runs and holiday rushes. And we found 66 nights where the company simply didn't have the cash. Even a perfect forecaster fails those, so we report them separately instead of blaming the model. Let me show you all of this live."

C Slide 6 — Results (1 min)

> "We tested seven strategies on the same year, hour by hour. Doing nothing: $41 million. The textbook Miller–Orr model: $25.5 million. Ballast: $20.3 million. That's half the cost of doing nothing and 20% better than the textbook. We even beat the best fixed buffer, which was picked *with hindsight* on the same year. And we are within 3% of an oracle that sees the future."

Point at the cost-of-confidence curve: "Each extra 'nine' of safety has a price. The treasurer chooses it with a slider, and we show exactly what it costs."

C Slide 7 — Trade-offs (1 min)

- **No bonds:** "Textbooks say to put some reserves in bonds for extra yield. We built it and tested it at five levels. Every level lost money this year, up to $766K, because bond prices moved more than the extra yield. So it's off by default, but still switchable."
- **Simple over complex:** "We chose a transparent formula over a black-box optimiser. A bank's risk team can check it on a spreadsheet, it decides in milliseconds, and a full year runs in about 6 seconds."
- **10× load:** "The work grows linearly with data. Ten times the data is about ten times the time, roughly a minute, and each night's decision stays instant."

C Slide 8 — Limits and next steps (1.5 min)

- "One year has only a handful of extreme nights, so our 99.5% buffer is close to the worst night we've ever seen. That's why real coverage came out at 96.5%, not 99.5%. Every miss was still paid. Next step: proper tail modelling (extreme value theory)."
- "The company loses $2.5 billion over the year. No sweep strategy fixes that. It's a capital problem, and we show it honestly instead of hiding it."
- "Only one sweep window, 5 pm. Next: intraday windows."
- "Production path: run in shadow mode beside the current process, then recommend, then automate with two-person approval for large sweeps."

A Close (1 min)

> "Three things to remember. One: half the cost, $21 million a year, with zero missed payouts. Two: every decision is explained night by night, and the maths fits on one line. Three: it's not a notebook, it's a live product. Upload any year of data and get this full analysis in under a minute. Thank you. We'd love your questions."

5 · Demo run-sheet (B drives, ~4.5 min)

Before you go on stage: open ballast.jkartik.in and click **Reference Dataset** once (the first run after a redeploy takes about 50 s, then it's instant). Have `data/synthetic_growing_bank.csv` on the desktop. Keep the printed report PDF open in another tab as a backup. Zoom the browser to 110–125% so the back row can read it. Turn off notifications.

CLICK	SAY
Home page → **Reference Dataset**	"This is the year of hourly data from the problem statement: 8,760 hours."
Scroll: **Deposits & withdrawals**, **Data quality**	"First it checks the file: no missing hours, no duplicates. Bad data would silently corrupt everything."
Trajectory of the book	"Notice this: the company loses $2.5 billion over the year. Keep it in mind; it matters later."
Overnight risk bar chart	"Friday and Saturday nights carry about five times the risk. That's why one fixed cushion fails."
Five steps at 17:00	"Here's the whole method with this dataset's own numbers: q* is 98.63%, buffers are $19M and $47M."
Decision log calendar (★ star moment)	"Every square is one night's decision. Green: covered. Red: the night beat the buffer, so we broke repo early and still paid everyone. Grey: the company simply ran out of capital." Click a square, then read **Why this call** aloud: "It explains every decision in plain English." Click **Next miss** and show that it was handled.
Risk appetite slider	Drag it. "A treasurer sets how safe they want to be and immediately sees what each extra level of safety costs per year."
Reserve calculator	Change the repo rate (e.g. 4 → 6). "If rates change, the buffer and the maths update live. Every step of the formula is shown."
Strategy comparison + **Net income**	"Five strategies on the same year. Ballast is gold; the green line is the perfect-foresight oracle, almost touching. All lines fall in November because the company runs out of cash, for every strategy alike."
Stress Lab	Pick **Bank run** and click **Run** (takes a few seconds). "We can inject a bank run on any date. Payouts during the run are funded; it shows how much liquidity is permanently lost."
Audit section	"These checks are re-run on every analysis: every dollar accounted for, sweeps only at 5 pm, and no look-ahead. We delete the future from the file, re-run, and get identical decisions."
Verdict + AI summary	"And the summary at the bottom is written by AI from the computed numbers only."
(If time) **Upload Data** → synthetic_growing_bank.csv	"It works on any data. Here's a different bank. Watch the charts appear as each stage finishes." (On this file Ballast beats the hindsight buffer by $176K.)

Print Report	“One click turns it into a board-ready PDF.”

If something breaks: say calmly “live demos, let me show you the saved run”, then open the report PDF. Never apologise more than once. Never debug in front of judges. Do **not** upload the volatile or bank-run synthetic files live: Ballast trails the hindsight baseline on those, which is fine to discuss but not to stumble into.

5b · Every section of the website, explained

In page order, top to bottom. For each section: **what it shows**, **how to read it**, and **the line to say**. The top bar has a row of links (Data, Risk, Method, Decisions, Calculator, Strategies, Backtest, Stress, Audit, Verdict) that jumps to any section.

Top bar buttons

Print Report turns the whole page into a clean white PDF that starts with the verdict. **Export** downloads the data: nightly decisions, strategy scoreboard, monthly P&L (CSV) or everything (JSON). ◐ switches light/dark. **New Analysis** goes back home. **Upload Data** analyses a new CSV.

> "Everything is exportable and printable. A treasurer can take this straight to a board meeting."

Home page

Drop a CSV or click **Reference Dataset** (the problem statement's year of data). Charts then appear one by one as each stage of the analysis finishes, instead of behind a long loading screen.

1. Deposits and withdrawals, hour by hour

Shows: the raw data, rolled up to one point per day. Green = money in, red = money out. Hover for the exact numbers. **Read:** the two lines are almost identical in size (about $240M a day each way), and the spikes are paydays and weekends. Withdrawals slightly exceed deposits: 51.4% of hours are net outflow.

> "This is the data exactly as provided: $89 billion in, $91 billion out over the year."

2. Data quality: "Checked before a single dollar is simulated"

Shows: ticks for row count (8,760 of 8,760), no missing or duplicate hours, no empty values, net flow consistent, every sweep window at 17:00 and maturity at 08:00, plus the average rates found in the file (repo ~4.06%, break fee 0.5%, haircuts 2%/15%, overdraft 12%). The small "validator flags" at the bottom are automatic warnings that we resolved; their explanations are in the assumptions register.

> "Garbage in, garbage out. So the engine validates every file before it trusts it."

3. Cash position: "The trajectory of the book"

Shows: cumulative deposits minus withdrawals across the year. **Read:** a steady slide to −$2.53B. The company is shrinking; it loses more money than it gains.

> "Remember this line. Withdrawals beat deposits by $2.5 billion. By November the cash runs out, for every strategy, even a perfect one. That's a capital problem, not a sweep problem."

4. Overnight risk: "Not every night carries the same risk"

Shows: the average worst overnight drain for each day of the week. Gold bars = Friday and Saturday. **Read:** Fri ≈ $27.8M and Sat ≈ $31.7M, against $4.6–7.9M for the rest.

> "Saturday nights are almost seven times riskier than Sunday nights. One cushion for every night is structurally wrong. This chart is why Ballast has two buffers."

5. Anomaly detection: "Spikes and depressions, explained"

Shows: the 14 biggest single-hour moves (dots above the line are inflows, below are outflows), and the likely cause of each: payday inflow (8), weekend spending (5), unclassified (1). Causes come from the calendar (day of month, day of week), not a hard-coded list of dates, so it works on any year.

> "The extremes aren't random. They're paydays and weekends, which is exactly what our weekend regime captures."

6. How Ballast decides: "Five steps at 17:00, every night"

Shows: the algorithm as five cards (Measure, Split, Price, Hold, Sweep), each with this dataset's real numbers in gold: 364 nights measured; weekend average $29.7M vs weekday $6.3M (4.7×); q* = 0.50% ÷ (0.50% + 0.0069%) = 98.63%; coverage 99.5% with the floor binding; buffers $19.0M and $46.7M; the waterfall order.

> "This is the entire strategy on one screen. A risk officer can check every number with a calculator."

7. Decision log: "Every night, every call — and why" ★ most important

Shows: a calendar with one square per night. Green = the buffer covered the night. Red = the night was worse than the buffer, so repo was broken early (payouts still made). Grey = the drain was bigger than all the cash the company had (capital-constrained, from November on). Pale green = warm-up nights in early January, when there was little history yet.

Click any square to see: free cash at 5 pm; the split bar (gold = held back, green = swept); buffer held, amount swept, the night's worst drain, spare or shortfall, and interest earned. The chart shows how withdrawals built up hour by hour from 17:00 to 08:00, against the gold dashed buffer line. If the line stays under the dashed line, the night was covered. **"Why this call"** explains the decision in words: which regime, how many past nights it looked at, the q* arithmetic, and what happened.

Buttons: Previous/Next, **Largest drain** (the worst night of the year), **Next miss** (jumps to a red night). Arrow keys also work.

Bottom, "96.5% of nights covered": the honest check. On nights where the company had enough cash, the buffer covered 272 of 282. The 10 misses were paid by breaking repo, with zero failed payouts. With one year of data, the 99.5% level is close to the worst night seen so far, so surprises cause misses.

> "Most teams show you a total. We show you all 364 decisions and the reason for each. Click any night."

8. Risk appetite: "What does the next 'nine' of confidence cost?"

Shows: a slider from 90% to 99.99% coverage. As you drag, it shows the weekday buffer, the weekend buffer and the yearly interest given up by holding that much cash. The curve underneath shows the cost rising steeply near 100%. The dashed line marks q* (98.63%).

> "Safety has a price. Going from 99.5% to 99.99% costs extra interest every year. The treasurer decides, and sees the price before deciding."

9. Reserve calculator: "Size tonight's reserve yourself"

Inputs: repo rate, early-break fee, lock window (hours), coverage floor and free cash at 5 pm. **Outputs:** the formula worked step by step (Cu = cost of being short, Co = cost of an idle dollar, q*, then q used), which one is binding (the floor or q*), a table of buffer, nights missed in this data and sweep amount for each regime, the yearly interest given up, and how much of that the safety floor adds over pure economics.

> "If interest rates go up, an idle dollar costs more, so you hold a smaller buffer. Type 6% and watch the buffer shrink. Same rule as the engine, to the dollar."

10. Strategy comparison: "Five approaches, one book"

Shows: the total cost of liquidity (lost interest plus all fees) for: all cash $41.3M, fixed $40M buffer $20.4M, moving average + 2σ $21.1M, **Ballast $20.3M (gold)**, oracle $19.8M (green). Lower is better. **View Detailed Metrics** opens a table: average balances, interest in bps, fees, overdraft hours, early breaks, approvals and payout rate.

> "Same data, same rules, five strategies. Ballast is the cheapest strategy you could actually deploy, and close to the impossible oracle."

Why overdraft hours are about 1,567 for everyone: the company runs out of capital in November whatever strategy it uses.

11. Backtest: "Net income, day by day" and "Month by month"

Chart: money earned minus fees, accumulated through the year, one line per strategy (gold = Ballast, green = oracle, red = all cash, which goes negative from overdraft fees). All lines rise until late October, then fall when the cash runs out. **Table:** Ballast's interest, fees and net profit each month, and the difference from the fixed buffer (green = Ballast ahead). Ballast loses January (−$110K) while it is learning with no history, then wins most months. The total is +$24K over the year.

> "The fixed buffer was picked by looking at the whole year in advance. Ballast starts blind in January, learns, and still finishes ahead."

12. Attribution: "The remaining distance to optimal"

Shows: why Ballast is $590K behind perfect. Bars: conservatism (cost of our 99.5% safety floor, about $5K), forecast error (surprising nights, about $585K) and total. On the right: 66 of 364 nights were capital-constrained, which no strategy could fix.

> "Almost all of the remaining gap is surprise nights, not our safety rule. That tells us exactly what to improve next: tail forecasting."

13. Stress testing: "Behavior under duress" and the Stress Lab

Top: four preset disasters, run against an emergency credit line of $1B: SVB-style bank run, holiday withdrawals ×3, payday ×2, bond yields +200bp. "Survived" means every payout was made all year. "First missed payout — date" means the shock drained so much cash that, combined with the company's yearly decline, the credit line ran out months later on that date.

Stress Lab: design your own shock (withdrawal surge, deposits stopping, bank run or payday surge), its start date, size and length, then click **Run**. You get a verdict, the liquidity permanently lost, the lowest point, extra fees, and a chart of cash with the shock (red) and without it (grey).

> "A judge can invent a disaster right now and we'll replay the whole year with it. During a 25% bank run, every payout during the run was still funded."

14. Audit: "Proven on this run, not asserted"

Six live checks, all PASS: **every dollar accounted for** (accounting balances every hour, within 50¢); **sweeps only at 17:00** (294 sweeps); **repo released only at 08:00** (55 early breaks, all logged emergencies); **no look-ahead** (future rows deleted, decisions identical); **every payout funded** (100%); **input data is clean**.

> “Don’t trust us. The app re-proves its own correctness every time it runs, even on a file uploaded a minute ago.”

15. Assumptions register: “Every judgment call, in the open”

Every decision the brief or the data left open, tagged **BRIEF** (from the problem statement), **DATA** (read from the file), **DESIGN** (our tested choice) or **ASSUMED** (unknown, so we parameterised it). Examples: $2B starting capital, 99.5% floor, bonds off, the hindsight-tuned baseline, the half-capital credit line in stress tests.

> “Every assumption is written down with its source, so nothing is hidden.”

16. The verdict (KPIs) and executive summary

Four headline numbers: net interest earned $21.35M, total cost of liquidity $20.35M, payout success 100%, optimal strategy: Ballast. Below them, a short AI-written summary built only from the computed figures. It sits at the bottom because writing it takes a few seconds, and it appears with a “sand-forming” text effect. The printed report puts this block first.

> “And the bottom line, summarised in plain English for the CFO.”

6 · Questions they will ask, and how to answer

Rule: answer in one sentence, give one number, then stop. If you don't know: "Good question. We didn't test that, and here's how we would." Honesty beats bluffing every time.

Q1. Why not just keep a fixed buffer, like $40M every night?

> Because risk changes by day: weekends are 5× riskier. A fixed buffer is too big on weekdays (wasted interest) or too small on weekends. And the best fixed buffer, $40M, could only be found by looking at the whole year in hindsight. In real life you can't do that. Ballast beat it anyway, with no hindsight.

Q2. You only beat the static buffer by $28K. Isn't that basically a tie?

> On this file the margin is small, yes. But that static buffer cheated: it was tuned on the same year it was scored on. Ballast starts with zero history and learns. On the growing-bank dataset Ballast wins by $176K. The big wins are against strategies you could actually deploy: 51% vs doing nothing, 20% vs Miller–Orr.

Q3. What is the newsvendor model, and why does it fit here?

> It answers "how much should I stock when demand is uncertain and being short costs more than being over?", like a newspaper seller choosing how many papers to print. Here, being short costs the 0.5% break fee and being over costs 15 hours of interest. Their ratio gives the percentile to cover: 98.6%.

Q4. Why measure the worst point of the night instead of the total?

> Because you go bankrupt at the lowest point, not at the end. If $30M leaves by 1 am and $10M comes back by 8 am, the total is $20M but you needed $30M. Sizing to the total under-covers exactly the nights that matter.

Q5. You target 99.5% coverage but got 96.5%. Isn't the model wrong?

> It's an honest limitation, and we show it in the app. With one year of data, the 99.5th percentile is basically the worst night seen so far, so new extreme nights cause misses. But every miss was paid by breaking repo; payouts stayed 100%. Our next step is extreme-value tail modelling to fix exactly this.

Q6. Is there any look-ahead / data leakage?

> No. Decisions only see data up to 4:59 pm that day, enforced in code. And the app proves it live: it deletes all future rows, re-runs, and gets identical decisions. It's in the Audit section.

Q7. Why do all strategies, even the oracle, go into overdraft?

> The data loses $2.53B over the year against a $2B starting balance. By November the money is gone no matter what you do. That's a capital problem; the fix is raising capital, not a better sweep. We separate those 66 nights so the model isn't blamed for them.

Q8. Where did the $2B starting balance come from?

> The dataset doesn't give one, so it's a documented assumption. It's large enough that strategy quality matters, and small enough that the year-end pressure stays visible. For uploaded files we scale it to 40× the worst night. It's in the assumptions register.

Q9. Why not invest in bonds for higher yield?

> We built it and tested it at 5%, 10%, 25% and 50%. Each one earned less, up to $766K worse, because bond prices fell more than the extra yield, and selling bonds early costs 2%. Data beat textbook, so bonds are off but switchable.

Q10. Why not machine learning / a more complex optimiser?

> Banks must explain their liquidity models to regulators. Our rule fits on one line and can be checked on a spreadsheet. It decides in milliseconds. With one year of data, a complex model would overfit. The forecaster feeding the rule is swappable, so ML can be plugged in later without changing the safety logic.

Q11. What happens under 10× load / more data?

> Runtime grows linearly: a year takes about 6 seconds, so ten years takes about a minute. Each nightly decision only looks at a recent window of about 100 days, so it stays instant however long the history gets.

Q12. What if a bank run happens?

> We test it in the Stress Lab. An SVB-style run pulls 25% of deposits in a day. Ballast funds every payout during the run through the waterfall and the emergency credit line. The app also shows how much liquidity is permanently lost and whether the line eventually runs out.

Q13. How do you handle a night much worse than history?

> The waterfall: break repo (0.5%), then sell bonds (2%), then stocks (15%), then overdraft (12% a year). A customer is never refused. It costs money, but the payout is made.

Q14. What is the oracle and why compare to it?

> A strategy that knows each night's drain in advance. It can't exist, but it tells us the best score possible. Ballast is within $598K, about 3%, of it. That's how we know there isn't much left to gain.

Q15. Where does the remaining gap to perfect come from?

> The Attribution section splits it: about 99% is forecast error (surprising nights), and almost nothing is from our safety floor. So the next improvement is better tail forecasting, not a different rule.

Q16. Why is the weekend Friday and Saturday, not Saturday and Sunday?

> Nights are named by the evening they start. Friday night and Saturday night are the ones with weekend spending. The data showed it: those two nights drain about 5× more. And it's recomputed from each uploaded file, not hardcoded.

Q17. Does it work on other data?

> Yes. Upload any CSV in the same format. Nothing is hardcoded to the sample year: spikes are classified from the calendar, and buffers come from that file's own history. (Offer to upload the growing-bank file.)

Q18. Is the AI summary making things up?

> No. It's only given the computed numbers and told to restate them. If the AI is unavailable, a template writes the same summary. The analysis itself uses no AI at all; it's all deterministic maths.

Q19. How would a real bank adopt this?

> Three phases: shadow mode (run beside the current process and compare), then recommendations a treasurer approves, then automation with two-person sign-off for sweeps over $2M. Every decision is logged and explainable.

Q20. What's novel here?

> Four things: (1) sizing to the worst point of the night, not the total; (2) separate weekday/weekend risk models; (3) pricing the buffer with the actual fee and interest rate; (4) a full product: night-by-night explanations, live audit proofs, stress lab and calculator, working on any uploaded file.

Q21. What would you do with another week?

> Extreme-value tail modelling to hit true 99.5% coverage; multiple sweep windows during the day; live interest-rate feeds; separate buffers per business line.

Q22. How do you know the accounting is right?

> Every hour, the app checks that cash plus all investments minus overdraft equals starting capital plus net flows plus income minus fees. It matches to within 50 cents across all 8,760 hours, and this runs live on every analysis.

Hard one you don't understand?

> "Let me make sure I answer the right question. Do you mean X or Y?" Then hand off: "*Name* worked on that part." Never guess a number.

7 · How to beat the other teams

Our edge, in the order judges care about:

1. **It's a real, live product**, not slides or a notebook. Let the judges use it on their phones (QR code). Most teams can't offer that.
2. **Explainability:** every one of 364 decisions has a plain-English "why". Judges can click any night. Competitors show totals; we show reasoning.
3. **Honest benchmarking:** we compare against a hindsight-tuned baseline *and* a perfect-foresight oracle, and we show our weaknesses (96.5%, capital-constrained nights). Judges trust teams that admit limits.
4. **Proof, not claims:** the Audit section re-proves no look-ahead and exact accounting live. Say: "Don't trust us, the app checks itself."
5. **Treasurer tools:** the risk slider, reserve calculator, stress lab and one-click PDF report show that real users were thought about.

Delivery tactics

- **Start with money, not maths.** "$41 million wasted" lands harder than "newsvendor fractile."
- **One number per sentence.** Pause after big numbers. Let "51%" and "zero missed payouts" sink in.
- **Use the analogies:** the newspaper seller (newsvendor); fuel for the worst stretch of a road trip (worst drain); the 292% trap.
- **Demo the decision log**. It is the most memorable moment. Click a red night, read the "why" aloud.
- **Own the weaknesses before they ask.** It turns their toughest question into your strength.
- **Finish on time.** Rehearse twice with a timer. Cut the 10× load and next-steps details first if you're running long.
- **Everyone should be able to give the one-breath pitch**, in case a judge asks the "wrong" person.

8 · Danger zone: fix or avoid these in the current deck

Some lines in "globalhost - Ballast - Final.pptx" don't match the live app. A judge who checks will lose trust. **Fix them in the slides, or simply don't say them:**

IN THE DECK	REALITY (SAY THIS INSTEAD)
Overdraft "costs 18% APR"	The data says **12% APR**.
"Average cash cushion under $33M"	The buffer is about $19M/$47M, but the *average cash balance* is about $335M (cash returns at 8 am and builds up during the day). Say "nightly buffer of $19M on weekdays, $47M on weekends."
Buffers "$15.0M weekday / $41.3M weekend"	The app shows **$19.0M / $46.7M** at 99.5%. Use the app's numbers.
"Tested a MILP via HiGHS, ~40 s per day"	Only say this if your team really ran it. Safer: "We chose a transparent closed-form rule over a black-box optimiser."

“Scales to 100,000 hours / 10 books in under 60 s”, “0.6 ms per decision”, “sub-100 ms ingest”	Not measured. Say “a full year runs in about 6 seconds; it grows linearly.”
Bond table “Total cost” column ($22.9M–$34.6M)	Only the **net profit** column ($14.58M → $13.82M) is verified. Point at that column only.
Slide 5 says the AI summary is at the top	It’s now at the bottom of the page. The page also has new sections (decision log, calculator, stress lab, audit). Re-take screenshots if there’s time.
“Gemini, zero hallucination”	Say “the AI only restates computed numbers, with a template fallback.” “Zero” invites a challenge.

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9 · Final checklist

- ☐ Hosted site redeployed (git pull + restart) and Reference Dataset clicked once to warm the cache.
- ☐ synthetic_growing_bank.csv on the desktop; report PDF open as backup; laptop charged; notifications off.
- ☐ Each speaker knows their slides, their hand-off line and the one-breath pitch.
- ☐ Two full timed rehearsals (target 14 minutes, leaving a buffer).
- ☐ Everyone has read the Q&A section aloud once.